YANG-TIAN SUN

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EDUCATION

The University of Hong Kong Oct 2023 – Present, Hong Kong

Ph.D. in Electrical and Electronic Engineering
Research interests: Video Generation and In-Context Learning; Generative 3D/4D Reconstruction; World/Action Models; Vision-Language-Action Models for Embodied AI and Computer Use
Supervisor: Prof. Xiaojuan Qi

University of Chinese Academy of Sciences Apr 2019 – Jun 2022, Beijing, China

M.S. in Computer Science GPA 3.81/4.0
Research interests: Computer graphics, 3D computer vision, deep learning
Supervisor: Prof. Lin Gao

University of Science and Technology Beijing Sep 2015 – Jun 2019, Beijing, China

B.S. in Applied Physics GPA 3.82/4.0 Ranking 1/86

PUBLICATIONS

* Equal contribution; † Project lead; # Corresponding author.

Publications (First Author / Co-First Author / Project Lead)

1. **MultiLive: On-Policy Distillation of Video and Multi-View Priors for Multi-View Video Generation**
   —- *"Distills video and multi-view priors on-policy to generate synchronized multi-view videos in four denoising steps, without multi-view video training data."*
   Yifan Niu*, **Yang-Tian Sun***,†, Haoran Feng, Lu Sheng, Yan-Pei Cao#, Zehuan Huang#,†.
   2026. [Paper]
2. **Mira-Scene: Pixel-Aligned Layouts for Generative 3D Scene Reconstruction**
   —- *"Reconstructs a coherent 3D scene from a single image using pixel-aligned layout correspondences."*
   Yang-Tian Sun*, Tianjia Liu*, Zehuan Huang, Yi-Hua Huang, Xiaoyang Lyu, Ziyi Yang, Zi-Xin Zou, Yuan-Chen Guo, Yan-Pei Cao#, Xiaojuan Qi#.
   arXiv:2609.23796, 2026.
3. **Stereo World Model: Camera-Guided Stereo Video Generation**
   —- *"Uses camera-guided stereo video generation to build geometrically consistent 3D-aware world models."*
   Yang-Tian Sun, Zehuan Huang, Yifan Niu, Lin Ma, Yan-Pei Cao, Yuewen Ma, Xiaojuan Qi.
   IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
4. **UniGeo: Taming Video Diffusion for Unified Consistent Geometry Estimation**
   —- *"Adapts video diffusion to estimate unified, temporally consistent scene geometry."*
   Yang-Tian Sun, Xin Yu, Zehuan Huang, Yi-Hua Huang, Yuan-Chen Guo, Ziyi Yang, Yan-Pei Cao, Xiaojuan Qi.
   arXiv:2505.24521, 2025.
5. **Splatter a Video: Video Gaussian Representation for Versatile Processing**
   —- *"Introduces a video Gaussian representation for versatile tracking, depth estimation, segmentation, and editing."*
   Yang-Tian Sun*, Yi-Hua Huang*, Lin Ma, Xiaoyang Lyu, Yan-Pei Cao, Xiaojuan Qi.
   Conference on Neural Information Processing Systems (NeurIPS), 2024, pp. 50401–50425.
6. **SC-GS: Sparse-Controlled Gaussian Splatting for Editable Dynamic Scenes**
   —- *"Discovers sparse object controls for editable dynamic-scene reconstruction and manipulation."*
   Yi-Hua Huang*, **Yang-Tian Sun***, Ziyi Yang*, Xiaoyang Lyu, Yan-Pei Cao#, Xiaojuan Qi#.
   IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.

7. **Robust Pose Transfer with Dynamic Details using Neural Video Rendering**
   —- *"Separates dynamic appearance details from geometry to improve robust human pose transfer."*
   Yang-Tian Sun, Hao-Zhi Huang, Xuan Wang, Yu-Kun Lai, Wei Liu, Lin Gao.
   IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(2), pp. 2660–2666, 2022.
8. **Human Motion Transfer with 3D Constraints and Detail Enhancement**
   —- *"Combines explicit 3D constraints and detail enhancement for high-fidelity human motion transfer."*
   Yang-Tian Sun, Qian-Cheng Fu, Yue-Ren Jiang, Zitao Liu, Yu-Kun Lai, Hongbo Fu, Lin Gao.
   IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(4), pp. 4682–4693, 2022.
9. **NeRF-Editing: Geometry Editing of Neural Radiance Fields**
   —- *"Uses shape priors to enable geometry editing of neural radiance fields without retraining."*
   Yu-Jie Yuan*, **Yang-Tian Sun***, Yu-Kun Lai, Yuewen Ma, Rongfei Jia, Lin Gao.
   IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

Publications (Co-Authored)

10. **Monocular Depth Estimation from a Single Image: Progress and Opportunities**.
    Muxin Liu, Xiaoyang Lyu, **Yang-Tian Sun**, Yihua Huang, Ziyi Yang, Peng Dai, Xiaojuan Qi.
    arXiv:2609.01172, 2026.
11. **Stabilizing Streaming Video Geometry via Dynamic Feature Normalization**.
    Xiaoyang Lyu, Mingjie Liu, Xiuzhe Wu, Rui Wang, Yihua Huang, **Yang-Tian Sun**, Shiyi Shi, Xiaojuan Qi.
    IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
12. **LaviGen: Repurposing 3D Generative Model for Autoregressive Layout Generation**.
    Haoran Feng*, Yifan Niu*, Zehuan Huang#, **Yang-Tian Sun**, Yuxin Peng, Lu Sheng#.
    IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
13. **VideoSSM: Autoregressive Long Video Generation with Hybrid State-Space Memory**.
    Yifei Yu, Xiaoshan Wu, Xinting Hu, Tao Hu, **Yang-Tian Sun**, Xiaoyang Lyu, Bo Wang, Lin Ma, Yuewen Ma, Zhongrui Wang, Xiaojuan Qi.
    arXiv:2512.04519, 2025.
14. **SeqTex: Generate Mesh Textures in Video Sequence**.
    Ze Yuan*, Xin Yu*,†, **Yang-Tian Sun**, Yuan-Chen Guo, Yan-Pei Cao, Ding Liang, Xiaojuan Qi#.
    SIGGRAPH Asia, 2025 (Conference Track).
15. **AnimaX: Animating the Inanimate in 3D with Joint Video-Pose Diffusion Models**.
    Zehuan Huang, Haoran Feng, **Yang-Tian Sun**, Yuan-Chen Guo†, Yan-Pei Cao#, Lu Sheng#.
    SIGGRAPH Asia, 2025 (Conference Track).
16. **Deformable Radial Kernel Splatting**.
    Yi-Hua Huang, MingXian Lin, **Yang-Tian Sun**, Ziyi Yang, Xiaoyang Lyu, Yan-Pei Cao#, Xiaojuan Qi#.
    IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
17. **Spec-Gaussian: Anisotropic View-Dependent Appearance for 3D Gaussian Splatting**.
    Ziyi Yang, Xinyu Gao, **Yang-Tian Sun**, Yi-Hua Huang, Xiaoyang Lyu, Wen Zhou, Shaohui Jiao, Xiaojuan Qi, Xiaogang Jin.
    Conference on Neural Information Processing Systems (NeurIPS), 2024, pp. 61192–61216.
18. **3DGSR: Implicit Surface Reconstruction with 3D Gaussian Splatting**.
    Xiaoyang Lyu, **Yang-Tian Sun**, Yi-Hua Huang, Xiuzhe Wu, Ziyi Yang, Yilun Chen, Jiangmiao Pang, Xiaojuan Qi.
    ACM Transactions on Graphics (TOG), 43(6), 2024 (SIGGRAPH Asia).
19. **3D-Aware Text-Driven Talking Avatar Generation**.
    Xiuzhe Wu, **Yang-Tian Sun**, Handi Chen, Hang Zhou, Jingdong Wang, Zhengzhe Liu, Xiaojuan Qi.
    European Conference on Computer Vision (ECCV), 2024, pp. 416–433.
20. **Total-Decom: Decomposed 3D Scene Reconstruction with Minimal Interaction**.
    Xiaoyang Lyu*, Chirui Chang*, Peng Dai, **Yang-Tian Sun**, Xiaojuan Qi.
    IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
21. **Interactive NeRF Geometry Editing with Shape Priors**.
    Yu-Jie Yuan, **Yang-Tian Sun**, Yu-Kun Lai, Yuewen Ma, Rongfei Jia, Leif Kobbelt, Lin Gao.
    IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 45(12), 2023.

Honours and Awards

- **National Scholarship**, 2015/2016 academic year
- **National Scholarship**, 2016/2017 academic year

Working Experience

VAST Jul 2023 – Present

Research Intern

- 3D/4D (Generative) Reconstruction, World Models, Real2Sim

NetEase Game AI Lab Jul 2022 – Sep 2023

AI Engineer

- Applied differentiable rendering to game asset production.

Alibaba Taobao Jul 2021 – Oct 2021

Research Intern

- Investigated deformation algorithms for neural radiance fields.

Tencent AI Lab Jun 2020 – Mar 2021

Research Intern

- Investigated dynamic-detail generation for neural human rendering.

Professional Services

Reviewer for SIGGRAPH, SIGGRAPH Asia, TPAMI, CVPR, ICCV, ECCV, ICLR, NeurIPS, TVCG.